

EDUCATION

Troy University, Troy, AL

Jan 2017 Dec 2020

Bachelor of Science in Computer Science (**Minors:** Physics, Mathematics)

- **Graduated:** Dec 2020 with 3.82 GPA
- **Honors:** Chancellor's List, Provost's List
- **Relevant Courses:** Intro to Object Oriented Programming, Data Structures, Analysis of Algorithms, Intro to Database Management, Machine Learning, Intro to Artificial Intelligence, Secure Software Development

WORK EXPERIENCE

- Architected and implemented a Migration service, successfully managing the ETL of millions of data rows across production DynamoDB tables, ensuring zero downtime and enhancing data reliability.
- Collaborated with client engineers to build a scalable credit card payment system using Microservices architecture, facilitating secure transaction processing and financial data management, significantly improving payment efficiency.
- Engineered an event-driven notification system enabling user subscriptions to price/inventory changes using AWS Serverless tools (Lambda, SNS, SQS, DynamoDB) and Nodejs (serverless, localstack) increasing the sales by \$30 millions in a quarter.
- Spearheaded the development of a backend system utilizing AWS serverless tools to seamlessly ingest and process millions of Kafka messages within the booking flow, facilitating the integration of add-on offerings during initial bookings and significantly boosting add-on sales while enhancing system scalability and performance.
- Tools Used: Java Spring, .NET, Typescript (React, Angular), Postgres, OAuth2, Jenkins, Docker, AWS, GCP, Flyway, Gradle, Terraform
- Implemented critical front-end features, such as image cropping and edit locking, enhancing the usability of Slalom's PEM website for 13000 internal users.
- Developed a Headless Drupal and React prototype to decouple CMS, improving site performance and flexibility for future development.
- Tools used: Drupal 8, PHP, Acquia Cloud, Twig, MySQL, JavaScript, Vue.js, React, Docker, Lando, Composer, HTML, CSS
- Designed and deployed a Chatbot for filing First Notices of Loss, streamlining claim processes and enhancing customer experience through intuitive interactions.
- Leveraged Microsoft's Language Understanding and Sentiment Analysis to refine Chatbot interactions, achieving a 50% increase in user engagement and satisfaction.
- Halved the notice filing time through Chatbot integration, significantly speeding up the claim initiation process during critical tests.
- Tools used: Microsoft Bot Framework SDK 4, Azure Cloud Services, LUIS AI, C#, JavaScript, HTML, CSS
- Provided personalized tutoring in STEM subjects to TRiO program participants, analyzed grades and student retention year over year, contributing to a notable improvement in their academic performance and retention rate.

PROJECTS

Project Hermes Python (pygame, pandas)

2nd Place @ Hack for Climate Solutions (Sep 2021)

- Engineered a traffic simulation using Python showcasing a 11% reduction in emissions and 4% increase in throughput via AI-driven traffic light optimization. Collaborated with Chicago Dept. of Transportation for real-world application planning. (<https://hackathon.slalom.com/teams/project-hermes>)

iBlink Python (OpenCV, imutils), Flask

1st Place @ TroyHack (Jan 2020)

- Developed an eye-controlled typing system with OpenCV, translating eye blinks to Morse code for text input, enhancing accessibility for users with limited mobility.

ConnectByBus Python, Flask, Selenium

@VandyHacks VI (Nov 2019)

- A website that returns shortest, fastest, and cheapest route from one city to another by connecting Megabus bus stations.

SKILLS

- **Certification:** AWS Solutions Architect – Associate (Issued March 2021), Salesforce Certified AI Associate (Issued Jan 2024)
- **Programming:** Java, Typescript, Python, C#, JavaScript, SQL
- **Web Development:** React, Angular, HTML/CSS, SQL, Twig
- **Cloud Computing:** AWS(EC2, Lambda, DynamoDB, API Gateway, CloudWatch, SQS, SNS, EC2, KMS, S3), GCP(GKE, Cloud Functions, Pub/Sub)
- **Tools & Frameworks:** Spring, Node, .NET, Drupal 8, Docker, Terraform, Postgres, Microsoft Bot Framework, GraphQL, Kafka
- **Soft Skills:** Project Management, Team Building, Communication, Critical Thinking, Empathy, Humor (*What's life without whimsy?*)

EXTRACURRICULARS

- 1st Place at Troy Open Chess Championship (Spring 2018), Troy Chess Club Vice-President (Fall 2018), Troy Nepalese Student Association Vice-President (2019), Hosted Intro to Web Scraping session at TroyHack (Jan 2018)
- **Interests:** Reading (www.goodreads.com/bishalniroula), Movies, Mountain biking, Cooking, Sitcoms, Hiking